

Sydney Girls High School



Mathematics Extension 2

YEAR 12

2023 HSC Assessment Task 3

Reading time:	5 minutes
Writing time:	60 minutes
Total marks:	50
Topics:	Complex Numbers II, Vectors and Integration

General Instructions:

- There are four (4) questions which are almost of equal value.
- Attempt all questions.
- Show all necessary working. Marks may be deducted for badly arranged work or incomplete working.
- Start each question on a new page.
- Write on one side of the paper only.
- Diagrams are NOT to scale.
- Board-approved calculators may be used.
- Write using a black or blue pen.
- White-out or correction tape is NOT to be used.
- A correct answer without working will be awarded a maximum of 1 mark.
- A Mathematics reference sheet is provided.

Student Name: _____ **Teacher Name:** _____

BLANK PAGE

Question 1**12 Marks**

a) Find $\int \frac{dx}{\sqrt{8-2x-x^2}}$. **2**

b)

i) Find the real numbers a , b and c such that $\frac{5}{x^2(2-x)} \equiv \frac{ax+b}{x^2} + \frac{c}{(2-x)}$. **2**

ii) Hence, or otherwise, find $\int \frac{20}{x^2(2-x)} dx$. **3**

c) Does the point $A(1, 0, 3)$ lie on the line $\tilde{r} = \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} + t \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix}$? **2**

Justify your answer.

d)

i) Express $\sin \theta$ in complex exponential form. **1**

ii) Hence, show that $\sin^3 \theta = \frac{1}{4}(3 \sin \theta - \sin 3\theta)$. **2**

Question 2 (Start writing on a NEW page)**13 Marks**

a)

i) Find $\int \frac{x^2}{1+x^2} dx$.

2

ii) Use integration by parts to evaluate $\int_0^1 2x \tan^{-1} x \, dx$.

2

b)

i) Find the values of z for which $z^6 = -64$ where $z = x + iy$. Give your answers in mod-arg form.

2

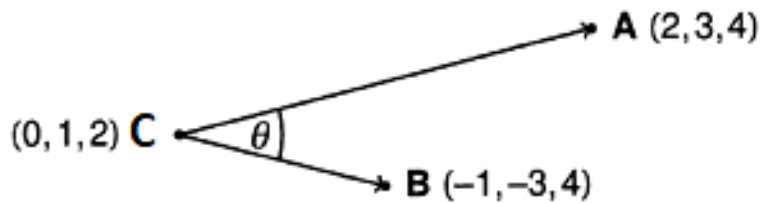
ii) Plot the solutions to part i) on an Argand diagram.

2

iii) The plotted roots are joined to form a polygon. Find the exact area of this polygon.

2

c) What is the shortest distance between the point **A** and a line passing through points **C** and **B**?

3

Question 3 (Start writing on a NEW page)**13 Marks**

a) Evaluate $\int_0^2 \sqrt{\frac{8-x}{x}} dx$.

3

b) A sphere $\mathcal{S}_1$ with centre $\underline{c} = 2\underline{i} + 2\underline{j} + 2\underline{k}$ passes through $\underline{a} = 4\underline{i} + 4\underline{j} + 4\underline{k}$.

i) Find the Cartesian equation of $\mathcal{S}_1$.

2

ii) A second sphere $\mathcal{S}_2$, has equation $(x-2)^2 + (y-2)^2 + (z-5)^2 = 1$.

3

Find the equation of the circle on which $\mathcal{S}_1$ and $\mathcal{S}_2$ intersect and state the centre and radius of this circle.

c) Given I_n , where n is an integer, $n \geq 0$:

$$I_n = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x dx$$

i) Show that $I_n + I_{n-2} = \frac{1}{n-1}$ for $n \geq 2$.

3

ii) Evaluate I_1 and hence determine the value of I_5 .

2

Question 4 (Start writing on a NEW page)

12 Marks

- a) Use the following definite integral property

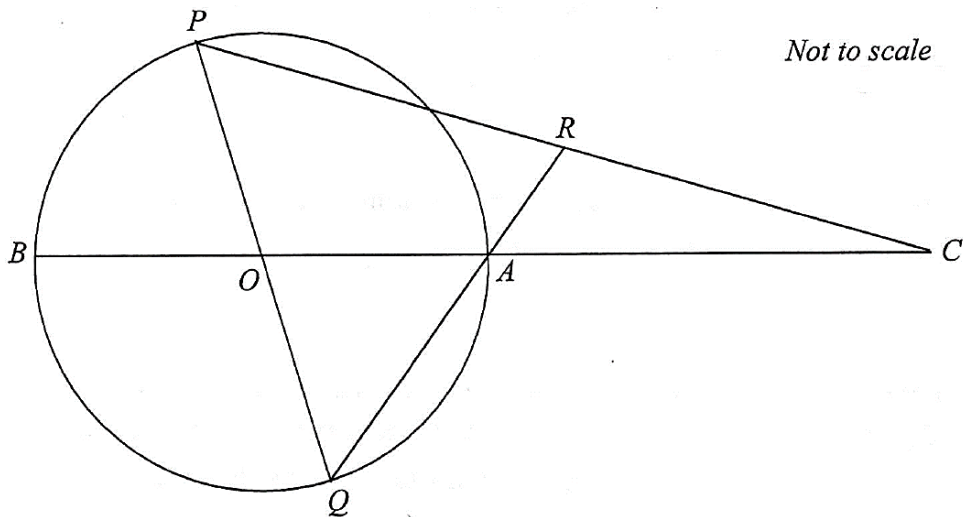
3

$$\int_{-a}^a f(x)dx = \int_0^a \{f(x) + f(-x)\}dx$$

to evaluate: $\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{e^x \sin^2 3x}{1+e^x} dx.$

- b)

4



In the diagram, PQ and AB are diameters of a circle with centre O .

BA is produced to C so that $BA = AC$. QA produced meets PC at R .

Let $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OP} = \underline{p}$.

Given $\overrightarrow{PR} = \lambda \overrightarrow{PC}$ and $\overrightarrow{QR} = \mu \overrightarrow{QA}$ for some scalars λ and μ , show that R is the midpoint of PC .

Question 4 continues on page 7

c) Consider the equation $z^5 = 1$.

i) Write down, in polar form, the five roots of $z^5 = 1$.

1

ii) Show that, for $z \neq 1$,

2

$$\frac{z^5 - 1}{z - 1} = \left(z^2 - 2z \cos\left(\frac{2\pi}{5}\right) + 1 \right) \left(z^2 - 2z \cos\left(\frac{4\pi}{5}\right) + 1 \right).$$

iii) Deduce that $\cos\left(\frac{2\pi}{5}\right)$ and $\cos\left(\frac{4\pi}{5}\right)$ are roots of the equation $4x^2 + 2x - 1 = 0$.

2

End of Task

Q1

$$a) \int \frac{dx}{\sqrt{8-2x-x^2}}$$

$$= \int \frac{dx}{\sqrt{9-(x+1)^2}} \quad \checkmark$$

$$= \sin^{-1}\left(\frac{x+1}{3}\right) + C \quad \checkmark$$

$$b) i) \frac{5}{x^2(2-x)} = \frac{ax+b}{x^2} + \frac{c}{2-x}$$

$$5 = (ax+b)(2-x) + cx^2 \quad \checkmark$$

$$\text{Coeff of } x^2: c - a = 0$$

$$\text{Coeff of } x: 2a - b = 0$$

$$\text{Constant}: 2b = 5 \rightarrow b = \frac{5}{2}$$

$$2a = b \rightarrow a = \frac{b}{2} = 5/4 \quad \checkmark$$

$$c = a \rightarrow c = 5/4$$

$$ii) \int \frac{20}{x^2(2-x)} dx = 4 \int \frac{5}{x^2(2-x)} dx$$

$$= 4 \left[\int \frac{\frac{5}{4}x + \frac{5}{2}}{x^2} dx + \int \frac{5/4}{(2-x)} dx \right] \quad \checkmark$$

$$= \int \frac{5}{x} dx + \int 10x^{-2} dx + \int \frac{5dx}{(2-x)} \quad \checkmark$$

$$= 5 \ln|x| - \frac{10}{x} - 5 \ln|2-x| + C \quad \checkmark$$

Q1 c) If $A(1, 0, 3)$ lies on

$$L = \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} + t \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix} \text{ then}$$

$$\begin{pmatrix} 1+2 \\ 0-3 \\ 3-5 \end{pmatrix} = t \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix} \text{ must be true for some } t. \checkmark$$

$$\begin{pmatrix} 3 \\ -3 \\ -2 \end{pmatrix} = t \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix} (*)$$

There is no values of t satisfy $(*)$ $\checkmark$
 $\therefore A(1, 0, 3)$ does not lie on L .

d) i) $e^{i\theta} = \cos\theta + i\sin\theta$ (1)

$$e^{-i\theta} = \cos(-\theta) + i\sin(-\theta)$$

$$e^{-i\theta} = \cos\theta - i\sin\theta$$
 (2)

$$(1) - (2) : e^{i\theta} - e^{-i\theta} = 2i\sin\theta$$

$$\sin\theta = \frac{1}{2i} (e^{i\theta} - e^{-i\theta}) \checkmark$$

Note: express $\sin\theta$ in the exponential form:

$$\therefore \sin\theta = \frac{1}{2i} (e^{i\theta} - e^{-i\theta})$$

Not $\sin\theta$ in terms of $\cos\theta$.

Q1

d(ii) from i) $\sin \theta = \frac{1}{2i} (e^{i\theta} - e^{-i\theta})$

$$\sin^3 \theta = \left[\frac{1}{2i} (e^{i\theta} - e^{-i\theta}) \right]^3$$

$$= \frac{-1}{8i} \left[e^{i3\theta} - 3e^{i\theta} + 3e^{-i\theta} - e^{-i3\theta} \right] \checkmark$$

$$= \frac{-1}{8i} \left[e^{i3\theta} - e^{-i3\theta} - 3(e^{i\theta} - e^{-i\theta}) \right]$$

$$= \frac{-1}{8i} \left[2i \sin 3\theta - 3(2i \sin \theta) \right]$$

$$= -\frac{1}{4} \sin 3\theta + \frac{3}{4} \sin \theta$$

$$= \frac{1}{4} (3 \sin \theta - \sin 3\theta) \checkmark$$

To get full marks, students must use the result from part (i) as the question stated "hence, show that: "

Question 2

2 (a) (i) $\int \frac{x^2 + 1 - 1}{x^2 + 1} dx = \int \left(\frac{x^2 + 1}{x^2 + 1} - \frac{1}{x^2 + 1} \right) dx \checkmark$

$$= \int \left(1 - \frac{1}{x^2 + 1} \right) dx$$
$$= x - \tan^{-1} x + C \checkmark$$

2 (ii) $\int_0^1 \underbrace{2x}_{v'} \underbrace{\tan^{-1} x}_u dx$

$v = x^2$
 $v' = 2x$
 $u = \tan^{-1} x$
 $u' = \frac{1}{x^2 + 1}$

$$= [uv]_0^1 - \int_0^1 u'v dx \checkmark$$
$$= [x^2 \tan^{-1} x]_0^1 - \int_0^1 \frac{x^2}{x^2 + 1} dx$$
$$= 1^2 \tan^{-1} 1 - \left[x - \tan^{-1} x \right]_0^1 \text{ (using (i))}$$
$$= \frac{\pi}{4} - \left(1 - \frac{\pi}{4} \right) \checkmark$$
$$= \frac{\pi}{2} - 1$$

2

(b) (i) Let $z = r \operatorname{cis} \theta$

$$\therefore (r \operatorname{cis} \theta)^6 = -64$$

$$r^6 \operatorname{cis} 6\theta = 64 \operatorname{cis} (\pi + 2\pi k) \quad \checkmark \text{ (de Moivre)}$$

Equate moduli and arguments:

$$r^6 = 64 \quad \text{and} \quad \text{For } k = -2, -1, 0, 1, 2, 3:$$

$$r = \sqrt[6]{64}$$

$$6\theta = \pi + 2\pi k$$

$$r = 2$$

$$\theta = \frac{\pi + 2\pi k}{6}$$

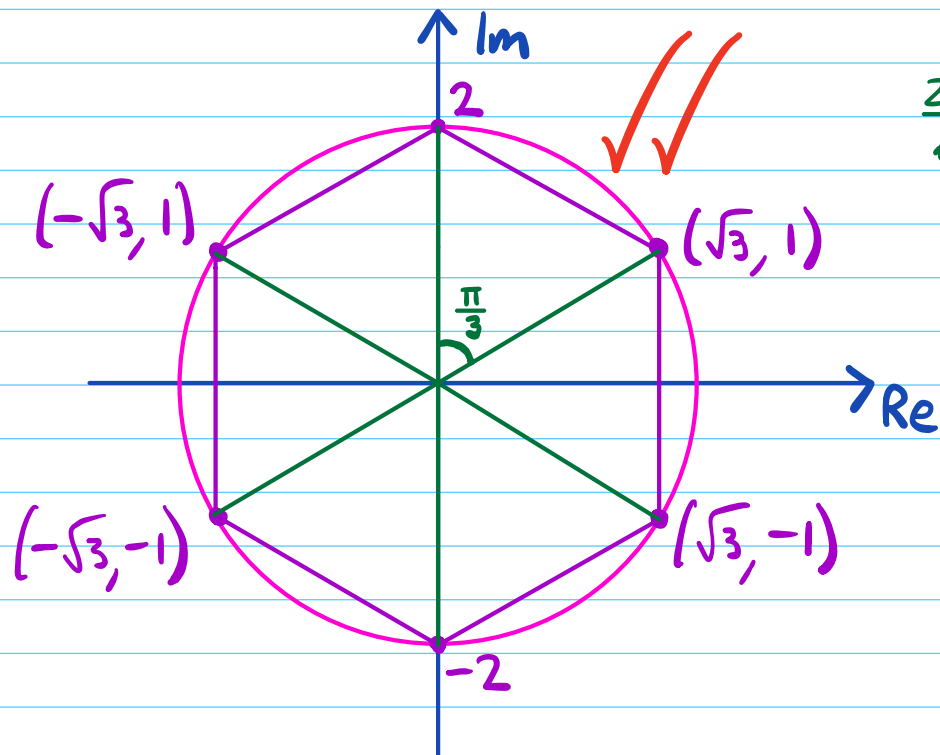
$$\theta = -\frac{\pi}{2}, -\frac{\pi}{6}, \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}$$

$$\therefore z = 2 \operatorname{cis} \left(\pm \frac{\pi}{2}\right), 2 \operatorname{cis} \left(\pm \frac{\pi}{6}\right), 2 \operatorname{cis} \frac{5\pi}{6}, 2 \operatorname{cis} \frac{7\pi}{6}$$

$$= \pm 2i, 2\left(\frac{\sqrt{3}}{2} \pm \frac{i}{2}\right), 2\left(-\frac{\sqrt{3}}{2} \pm \frac{i}{2}\right) \quad \checkmark$$

$$= \pm 2i, \sqrt{3} \pm i, -\sqrt{3} \pm i$$

2 (ii)



$$\frac{2\pi}{6} = \frac{\pi}{3}$$

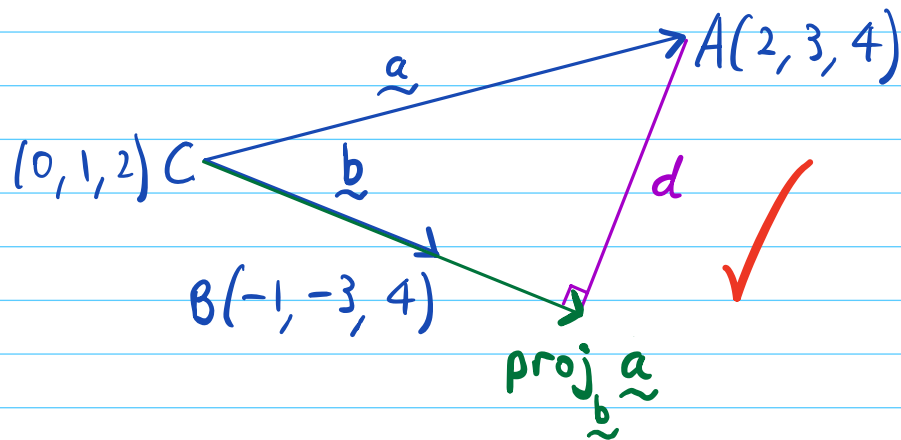
2 (iii) Construct radii as shown above. (green)

This creates six congruent equilateral triangles. ✓ The exact area of the hexagon is therefore

$$\begin{aligned} 6 \times \text{Area } \triangle &= 6 \times \frac{1}{2} ab \sin C \\ &= 6 \times \frac{1}{2} (2)(2) \sin \frac{\pi}{3} \\ &= 12 \left(\frac{\sqrt{3}}{2} \right) \checkmark \\ &= 6\sqrt{3} \text{ square units} \end{aligned}$$

3

(c)



The shortest distance between a point and a line is the perpendicular distance, so we first find the projection of $\vec{CA}$ onto $\vec{CB}$:

$$\text{proj}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{\vec{b} \cdot \vec{b}} \vec{b}$$

$$= \frac{\begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -4 \\ 2 \end{pmatrix}}{\begin{pmatrix} -1 \\ -4 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -4 \\ 2 \end{pmatrix}} \begin{pmatrix} -1 \\ -4 \\ 2 \end{pmatrix}$$

$$= \frac{-2 - 8 + 4}{1 + 16 + 4} \begin{pmatrix} -1 \\ -4 \\ 2 \end{pmatrix}$$

$$= -\frac{2}{7} \begin{pmatrix} -1 \\ -4 \\ 2 \end{pmatrix}$$

Note:

$$\vec{a} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$$

$$\vec{b} = \begin{pmatrix} -1 \\ -3 \\ 4 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} -1 \\ -4 \\ 2 \end{pmatrix}$$

$$\therefore d = \left| \vec{a} - \text{proj}_{\vec{b}} \vec{a} \right|$$

$$= \left| \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} - -\frac{2}{7} \begin{pmatrix} -1 \\ -4 \\ 2 \end{pmatrix} \right|$$

$$= \frac{6\sqrt{14}}{7} \text{ units}$$

a) Evaluate $\int_0^2 \sqrt{\frac{8-x}{x}} dx$.

3

$$= \int_0^2 \frac{\sqrt{8-x}}{\sqrt{x}} \times \frac{\sqrt{8-x}}{\sqrt{8-x}} \quad \text{"rationalise the numerator"}$$

$$= \int_0^2 \frac{8-x}{\sqrt{8x-x^2}} \quad \checkmark$$

$$= \int_0^2 \frac{4-x}{\sqrt{8x-x^2}} + \frac{4}{\sqrt{8x-x^2}}$$

$$= \int_0^2 (4-x)(8x-x^2)^{-\frac{1}{2}} dx + \int_0^2 \frac{4}{\sqrt{16-(x-4)^2}} dx$$

$$= \int_0^2 \frac{1}{2}(8-2x)(8x-x^2)^{-\frac{1}{2}} dx + \int_0^2 \frac{4}{\sqrt{16-(x-4)^2}} dx$$

$$= \left[\sqrt{8x-x^2} \right]_0^2 + \left[4 \sin^{-1} \left(\frac{x-4}{4} \right) \right]_0^2 \quad \checkmark$$

$$= \sqrt{12} + 4 \left(-\frac{\pi}{6} + \frac{\pi}{2} \right)$$

$$= 2\sqrt{3} + \frac{4\pi}{3} \quad \checkmark \quad (3)$$

Comment:

- This question was marked generously.
- Half the students didn't realise that they needed to 'rationalise the numerator' as the first step.
- Refer to the SGHS booklet examples for revision.

b) A sphere S_1 with centre $c = 2i + 2j + 2k$ passes through $a = 4i + 4j + 4k$.

i) Find the Cartesian equation of S_1 .

2

ii) A second sphere S_2 , has equation $(x-2)^2 + (y-2)^2 + (z-5)^2 = 1$.

3

Find the equation of the circle on which S_1 and S_2 intersect and state the centre and radius of this circle.

i) The radius of the sphere is given by:

$$\begin{aligned} r &= |a - c| \\ &= \sqrt{2^2 + 2^2 + 2^2} \\ &= \sqrt{12} \end{aligned}$$

$\therefore$ The cartesian equation of S_1 :

$$(x-2)^2 + (y-2)^2 + (z-2)^2 = 12$$

ii) Solving simultaneously:

$$(x-2)^2 + (y-2)^2 + (z-2)^2 = 12$$

$$(x-2)^2 + (y-2)^2 + (z-5)^2 = 1$$

Subtracting equation S_2 from S_1

$$(z-2)^2 - (z-5)^2 = 12 - 1$$

$$z^2 - 4z + 4 - (z^2 - 10z + 25) = 11$$

$$z^2 - 4z + 4 - z^2 + 10z - 25 = 11$$

$$6z - 21 = 11$$

$$6z = 32$$

$$z = \frac{32}{6} = \frac{16}{3}$$

✓ one mark for process of solving simultaneous equations.

Substitute $z = \frac{16}{3}$ into S_2 :

$$(x-2)^2 + (y-2)^2 + \left(\frac{16}{3} - 2\right)^2 = 12$$

$$\therefore (x-2)^2 + (y-2)^2 = \frac{8}{9}$$

one mark for eqn of circle.

✓ which is the equation of the circle.

The circle has centre $= (2, 2, \frac{16}{3})$ and radius $r = \frac{\sqrt{8}}{3}$.

✓ one mark for centre.

Comment:

* Many students lost marks because they made errors when solving simultaneous equations.

* Students needed to show a 2-D equation for the circle and centre $= (x, y, z) = (2, 2, \frac{16}{3})$.

c) Given I_n , where n is an integer, $n \geq 0$:

$$I_n = \int_{\pi/4}^{\pi/2} \cot^n x \, dx$$

i) Show that $I_n + I_{n-2} = \frac{1}{n-1}$ for $n \geq 2$. 3

ii) Evaluate I_1 and hence determine the value of I_5 . 2

$$i) I_n = \int_{\pi/4}^{\pi/2} \cot^n x \, dx$$

$$= \int_{\pi/4}^{\pi/2} \cot^2 x \cot^{n-2} x \, dx \quad \checkmark$$

$$= \int_{\pi/4}^{\pi/2} (\operatorname{cosec}^2 x - 1) \cot^{n-2} x \, dx$$

$$= \int_{\pi/4}^{\pi/2} \operatorname{cosec}^2 x \cot^{n-2} x \, dx - \int_{\pi/4}^{\pi/2} \cot^{n-2} x \, dx$$

$$= \left[-\frac{\cot^{n-1} x}{n-1} \right]_{\pi/4}^{\pi/2} - I_{n-2} \quad \checkmark$$

$$\therefore I_n + I_{n-2} = \frac{\cot^{n-1} \frac{\pi}{4}}{n-1} - \frac{\cot^{n-1} \frac{\pi}{2}}{n-1}$$

$$\therefore I_n + I_{n-2} = \frac{1}{n-1} \quad \text{for } n \geq 2.$$

Comment:
This was the simplest approach to answer this question. Refer to SLTTS booklet for revision.

Carefully shown last step.

$$ii) I_1 = \int_{\pi/4}^{\pi/2} \cot x \, dx$$

$$= \int_{\pi/4}^{\pi/2} \frac{\cos x}{\sin x} \, dx$$

$$= \left[\ln |\sin x| \right]_{\pi/4}^{\pi/2}$$

$$= \ln \left| \sin \frac{\pi}{2} \right| - \ln \left| \sin \frac{\pi}{4} \right|$$

$$= \ln 1 - \ln \frac{1}{\sqrt{2}}$$

$$= \frac{1}{2} \ln 2. \quad (\text{or } \ln \sqrt{2}) \quad \checkmark$$

$$I_5 + I_3 = \frac{1}{4} \quad (n=5)$$

$$I_5 = \frac{1}{4} - I_3$$

$$= \frac{1}{4} - \left(\frac{1}{2} - I_1 \right) \quad (n=3)$$

$$= -\frac{1}{4} + I_1$$

$$= -\frac{1}{4} + \frac{1}{2} \ln 2 \quad \checkmark$$

* Many students made errors evaluating I_1 and hence I_5
* Carry errors were awarded ① mark.

Question 4

$$\begin{aligned} (a) \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{e^x \sin^2 3x}{1+e^x} dx &= \int_0^{\frac{\pi}{3}} \frac{e^x \sin^2 3x}{1+e^x} + \frac{e^{-x} \sin^2(-3x)}{1+e^{-x}} dx \quad \checkmark \\ &= \int_0^{\frac{\pi}{3}} \frac{e^x \sin^2 3x}{1+e^x} + \frac{\sin^2 3x}{e^x + 1} dx \\ &= \int_0^{\frac{\pi}{3}} \frac{\sin^2 3x (e^x + 1)}{1+e^x} dx \\ &= \int_0^{\frac{\pi}{3}} \sin^2 3x dx \quad \checkmark \\ &= \frac{1}{2} \int_0^{\frac{\pi}{3}} 1 - \cos 6x dx \\ &= \frac{1}{2} \left[x - \frac{1}{6} \sin 6x \right]_0^{\frac{\pi}{3}} \\ &= \frac{1}{2} \left(\frac{\pi}{3} - \frac{1}{6} \sin 2\pi - 0 \right) \\ &= \frac{\pi}{6} \quad \checkmark \end{aligned}$$

Comment

Most students were successful in applying the given property.

However, many students had difficulty in going

from $\frac{e^{-x} \sin^2(-3x)}{1+e^{-x}}$ to $\frac{\sin^2 3x}{e^x + 1}$ $\times \frac{e^x}{e^x}$

$\left(\text{Note: } \sin^2(-3x) = (\sin(-3x))^2 = (-\sin 3x)^2 = \sin^2 3x \right)$

$$\begin{aligned}
 (b) \quad \vec{PR} &= \lambda \vec{PC} \\
 &= \lambda (\vec{OC} - \vec{OP}) \\
 &= \lambda (3\vec{a} - \vec{p})
 \end{aligned}$$

✓ Finds $\vec{PR}$ (or equivalent merit)

Also,

$$\begin{aligned}
 \vec{PR} &= \vec{QR} - \vec{QP} \\
 &= \mu \vec{QA} - 2\vec{p} \\
 &= \mu (\vec{p} + \vec{a}) - 2\vec{p}
 \end{aligned}$$

✓ Finds another expression for $\vec{PR}$

(or another expression for the previously obtained vector)

$$\begin{aligned}
 \therefore \lambda (3\vec{a} - \vec{p}) &= \mu (\vec{p} + \vec{a}) - 2\vec{p} \\
 3\lambda \vec{a} - \lambda \vec{p} &= \mu \vec{a} + (\mu - 2)\vec{p}
 \end{aligned}$$

$$\begin{cases} 3\lambda = \mu \\ -\lambda = \mu - 2 \end{cases}$$

$$-\lambda = 3\lambda - 2$$

$$\lambda = \frac{1}{2}$$

$$\begin{aligned}
 \therefore \vec{PR} &= \lambda \vec{PC} \\
 &= \frac{1}{2} \vec{PC}
 \end{aligned}$$

ie R is the midpt of PC.

Comment Poorly done.

(c)

$$\begin{aligned} \text{(i)} \quad z^5 &= 1 \\ &= \text{cis}(0 + 2k\pi) \\ &= \text{cis}(2k\pi) \end{aligned}$$

$$z = \text{cis}\left(\frac{2k\pi}{5}\right), \quad k = 0, \pm 1, \pm 2$$

$$= \underset{\substack{\uparrow \\ 1}}{\text{cis } 0}, \text{cis}\left(\pm \frac{2\pi}{5}\right), \text{cis}\left(\pm \frac{4\pi}{5}\right) \quad \checkmark$$

$$\begin{aligned} \text{(ii)} \quad z^5 - 1 &= (z-1)\left(z - \text{cis}\frac{2\pi}{5}\right)\left(z - \text{cis}\left(-\frac{2\pi}{5}\right)\right)\left(z - \text{cis}\frac{4\pi}{5}\right)\left(z - \text{cis}\left(-\frac{4\pi}{5}\right)\right) \quad \checkmark \\ &= (z-1)\left(z^2 - 2\cos\frac{2\pi}{5}z + 1\right)\left(z^2 - 2\cos\frac{4\pi}{5}z + 1\right) \end{aligned}$$

$$\therefore \frac{z^5 - 1}{z - 1} = \left(z^2 - 2\cos\frac{2\pi}{5}z + 1\right)\left(z^2 - 2\cos\frac{4\pi}{5}z + 1\right) \quad \checkmark$$

$$\text{(iii)} \quad z^5 - 1 = (z-1)(1 + z + z^2 + z^3 + z^4)$$

$\therefore$ From (ii),

$$1 + z + z^2 + z^3 + z^4 = \left(z^2 - 2\cos\frac{2\pi}{5}z + 1\right)\left(z^2 - 2\cos\frac{4\pi}{5}z + 1\right)$$

Equate coeff of z :

$$1 = -2\cos\frac{2\pi}{5} - 2\cos\frac{4\pi}{5}$$

$$\therefore \cos\frac{2\pi}{5} + \cos\frac{4\pi}{5} = -\frac{1}{2} \quad \checkmark$$

Equate coeff of z^2 :

$$1 = 1 + 4\cos\frac{2\pi}{5}\cos\frac{4\pi}{5} + 1$$

$$\therefore \cos\frac{2\pi}{5}\cos\frac{4\pi}{5} = -\frac{1}{4}$$

$\therefore \cos\frac{2\pi}{5}$ and $\cos\frac{4\pi}{5}$ are roots of

$$x^2 + \frac{1}{2}x - \frac{1}{4} = 0$$

ie. $4x^2 + 2x - 1 = 0$

Comment

Part (iii) was poorly done.

Students were asked to deduce (ie use previous result(s)).

Hence, other methods (such as substitution or using the quadratic formula and exact values for $\cos\frac{2\pi}{5}$ and $\cos\frac{4\pi}{5}$) were not awarded any marks.